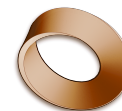


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Design



1.C – Building the Interactive Card

Unit 1: Interactive Animations & Games

Learning intentions

- Plan your card in enough detail that the coding becomes the easy part
- Build a working interactive card in Game Lab
- Balance how it looks against what it does, and know which you are playing to

Theory

This is **Criterion C: Creating the solution** — the build. You have a specification and a chosen design; now you make it real in Game Lab.

Plan first. The single strongest predictor of a good grade here is the detail in your plan, because a vague plan turns into a coding session where you are inventing and debugging at the same time. Work out what your sprites are, what changes about them, and what the user can do, **before** you open the editor.

The target your card has to hit. Your card should include at minimum:

- **3 sprites** with updating properties — using both random movement and the counter pattern
- **2 conditionals** that respond to user input, such as a key press or a mouse click
- **1 conditional** triggered by a variable or a sprite property, such as the sprite going past a certain position
- some **text**

The commands you have. For shapes: `background(color)`, `rect(x, y, width, height)`, `ellipse(x, y, width, height)`, `line(x1, y1, x2, y2)`, `text(string, x, y, width, height)`, `textSize(pixels)`. For colour and style: `fill('color')`, `noFill()`, `stroke('color')`, `noStroke()`, `strokeWeight()`.

One trap worth knowing. A sprite holds a whole collection of variables inside it. If you write `s = createSprite(...)` more than once, you create several sprites but the variable `s` only ever points at the newest one — the earlier sprites are still on screen but you can no longer control them. Give each sprite its own variable name.

How you are graded. Three things: how **attractive** the card is, how well it **functions**, and how **technically complex** it is. You will not have time to make something both hugely complex and completely polished. Decide which is your strength, play to it, and make sure the other parts still work.

You may ask your teacher or your classmates for help, but they may not do the project for you. Good coders rarely work alone.

Grading for the plan (Strand i/iv):

- 1-2 marks: limited plan
- 3-4 marks: simple plan
- 5-6 marks: reasonably detailed plan
- 7-8 marks: detailed plan

Grading for the build (Strand ii/iii):

- 1-2 marks: limited technical skills; something drawn but no interactive features
- 3-4 marks: basic interactive features which may not function fully. A card with limited technical skill but high-quality hand-drawn art fits here
- 5-6 marks: a complete interactive card with a number of interactive features, and/or a high standard of hand-drawn art, and/or other complex technical features
- 7-8 marks: as 5-6, plus some or all of exceptional graphics made by you, advanced interactive features that are not simply copied from the examples, or other advanced features suited to the card

Activity

Work through the planning questions below, then build the card in Game Lab using **Lesson 18** of the code.org unit.

If you finish before the others, keep going into the later lessons — velocity, collision detection and the side scroller are where cards start turning into games.

Questions

1. Describe your scene in a few sentences. What does the user see when the card opens, and what happens next?

2. Sprite 1 — label, what the image looks like, and which properties update in the draw loop.

3. Sprite 2 — label, what the image looks like, and which properties update in the draw loop.

4. Sprite 3 — label, what the image looks like, and which properties update in the draw loop.

5. Which sprite uses random movement, and which uses the counter pattern? Write the line of code you expect to use for each.

6. Conditional 1 responding to user input — what is the condition, and what happens as a result?

7. Conditional 2 responding to user input — what is the condition, and what happens as a result?

8. Conditional 3, triggered by a variable or a sprite property — what is the condition, and what happens as a result?

9. What variables do you need to store information, and what does each one hold?

10. What text appears on the card, and where on the 400 by 400 screen does it go?

11. Are you playing to attractiveness or to technical complexity? Say which, and what that means you will spend your time on.

Extension

Add something to the card that was not in your plan and is not in the example code — a second screen, a sprite that appears only after several clicks, a sprite that changes colour on a timer, or hand-drawn artwork of your own. Write down what you added and how you worked out how to do it. Features you worked out yourself are what separate the top band from the rest.
